

Muhammad Umar Yaksambi

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EDUCATION

RV College of Engineering

BE, Computer Science and Engineering (Data Sciences) – CGPA: 9.12/10

Bengaluru, KA

Expected May 2027

EXPERIENCE

HPE CPP3 Intern

Hewlett Packard Enterprise

March 2026 – Present

Remote

- Built a hybrid defect-mapping engine combining a 50/50 lexical and semantic scoring model with BAAI/bge-small-en-v1.5 embeddings to link CI failures to Jira issues, extending it with a 0.90 cosine-similarity duplicate-detection layer within 7-day proximity windows
- Optimized concurrent read/write database performance by implementing SQLite Write-Ahead Logging (WAL) and a 32MB cache extension, eliminating dashboard lockups during active pipeline ingestion
- Implemented a fault-tolerant REST API integration layer with idempotent write-backs and exponential backoff retry logic to safely interact with Atlassian's systems under rate limits

Product Developer Intern

SpikedAI

June 2025 – March 2026

Remote

- Architected event-driven backend services using FastAPI and Node.js, achieving a sustained 500 RPS throughput to support 500+ concurrent users for a sales meeting intelligence platform
- Optimized RAG and GraphQL pipelines to deliver a 750ms end-to-end response time against an indexed knowledge base of 100M+ tokens, scaling system throughput to 1 million tokens per minute
- Engineered multi-cloud infrastructure across AWS and GCP, orchestrating CI/CD pipelines via GitHub Actions and Nginx load balancing to ensure 99.9% service uptime in production
- Built responsive user interfaces leveraging React, TypeScript, and Tailwind CSS, integrating Redis caching to accelerate the delivery of actionable sales insights generated from Google Meet transcripts

PRISM Research Intern

Samsung R&D Institute

May 2025 – January 2026

Bengaluru, KA

- Built a real-time volumetric rendering pipeline for a 3D Gaussian Splatting PLY viewer using the Filament engine, achieving 30 FPS at 1080p on Dimensity 9000 mobile hardware with dynamic lighting support, replacing computationally heavy traditional ray-marching
- Optimized spherical harmonics processing and matc material compilation to render large-scale scenes comprising over 3 million Gaussians on mobile hardware

PROJECTS

Adaptive Defenses for Wireless IoT Networks | Python, Game Theory, Q-Learning

- Developed and simulated dynamic defense strategies for IoT networks using game-theoretic models and Q-learning to counter jamming attacks, achieving a 7% performance improvement over static baselines

Federated Learning with DP for Healthcare Recommendations | Flower, TFF, PyTorch, Llama-7B

- Built a privacy-preserving healthcare chatbot using federated learning and differential privacy, retaining patient history locally while continuously improving a global Llama-7B quantized model

Resilient Disaster Communication Framework | Embedded C, ESP32, ESP-NOW

- Designed an ESP32-based opportunistic mesh and mobile relay system for infrastructure-independent disaster communication, integrating 5 sensors for proactive environmental hazard prediction

LEADERSHIP AND ACTIVITIES

Vice President

The Coding Club of RVCE

February 2025 – Present

Bengaluru, KA

- Organized events and workshops for 200+ students, mentored juniors in C++, Python, and ML, and led sessions on APIs, LLMs, and competitive programming at RVITM and RVCE

SKILLS

Languages: Python, C++, TypeScript, SQL, Embedded C

Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, Node.js, React, Next.js, LangChain, GraphQL

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, GitHub Actions, Nginx

Databases: PostgreSQL, MongoDB, Redis, Qdrant, ChromaDB